

Passwordless Authentication Between Two Linux Servers & PuTTY Login

Scenario

Server	IP Address
Server 1	192.168.0.117
Server 2	192.168.0.118

Goal:

- Login from `.117` → `.118` without password
- Login using PuTTY without password

PART 1 — Passwordless Authentication Between Linux Servers

Step 1: Install SSH Packages

Run on BOTH servers:

```
sudo dnf install openssh-server openssh-clients -y
```

Step 2: Start SSH Service

Run on BOTH servers:

```
sudo systemctl start sshd  
sudo systemctl enable sshd
```

Verify:

```
systemctl status sshd
```

Step 3: Generate SSH Key

Run on source server (`192.168.0.117`):

```
ssh-keygen
```

Press:

- Enter
- Enter
- Enter

SSH keys will be created inside:

```
~/.ssh/
```

Files:

File	Purpose
id_rsa	Private Key
id_rsa.pub	Public Key

Step 4: Copy Public Key to Destination Server

Run from `.117` :

```
ssh-copy-id root@192.168.0.118
```

Enter destination server password ONE time.

Expected output:

```
Number of key(s) added: 1
```

Step 5: Test Passwordless Login

Run:

```
ssh root@192.168.0.118
```

If successful:

- Login happens without password
-

Manual Method (If ssh-copy-id Doesn't Work)

Step 1: Copy Public Key

Run on source server:

```
cat ~/.ssh/id_rsa.pub
```

Copy entire output.

Step 2: Paste Key on Destination Server

Run on destination server:

```
mkdir -p ~/.ssh  
nano ~/.ssh/authorized_keys
```

Paste copied public key.

Save and exit:

- Ctrl + O
 - Enter
 - Ctrl + X
-

Step 3: Fix Permissions

Run on destination server:

```
chmod 700 ~/.ssh  
chmod 600 ~/.ssh/authorized_keys
```

Step 4: Configure SSH

Open SSH config:

```
nano /etc/ssh/sshd_config
```

Ensure these lines exist:

```
PubkeyAuthentication yes  
PermitRootLogin yes  
AuthorizedKeysFile .ssh/authorized_keys
```

Step 5: Restart SSH Service

```
systemctl restart sshd
```

Step 6: SELinux Fix (CentOS Important)

```
restorecon -Rv ~/.ssh
```

PART 2 — PuTTY Passwordless Login

Step 1: Install PuTTY & PuTTYgen

Download:

- PuTTY
 - PuTTYgen
-

Step 2: Copy Private Key to Windows

Run on Linux:

```
cat ~/.ssh/id_rsa
```

Copy entire content.

Save in Windows as:

```
id_rsa
```

Step 3: Convert Key Using PuTTYgen

Open:

- PuTTYgen

Click:

```
Conversions → Import Key
```

Select:

- id_rsa
-

Step 4: Save `.ppk` File

Click:

Save private key

Save as:

mykey.ppk

Step 5: Configure PuTTY

Open PuTTY.

Session

Host Name:

192.168.0.118

Connection → Data

Auto-login username:

root

Connection → SSH → Auth → Credentials

Browse and select:

mykey.ppk

Step 6: Open Connection

Click:

Open

Login will happen without password.

Troubleshooting

Error: "Server refused our key"

Run on destination server:

```
chmod 700 ~/.ssh  
chmod 600 ~/.ssh/authorized_keys  
restorecon -Rv ~/.ssh  
systemctl restart sshd
```

Check SSH Service

```
systemctl status sshd
```

Check Firewall

```
firewall-cmd --permanent --add-service=ssh  
firewall-cmd --reload
```

Important Security Notes

✗ Never share:

- id_rsa
- .ppk files

✓ Only share:

- id_rsa.pub

Quick Commands Summary

Linux to Linux

```
ssh-keygen  
ssh-copy-id root@192.168.0.118  
ssh root@192.168.0.118
```

PuTTY

```
Import id_rsa → Save .ppk → Load in PuTTY
```

Real DevOps Usage

This setup is commonly used in:

- Jenkins deployments
- Ansible automation
- CI/CD pipelines
- SCP file transfers
- Remote server management